CPE 381 Classwork 08: Continuous-Time Signals, Linearity & Superposition

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: September 16, 2026 (In-Class)

1. Check if the signal $x(t) = e^{-at}u(t), \quad a > 0$ is an energy signal. Note: a signal is energy signal if its energy is finite. **(10 Points)**

Answer

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |e^{-at}u(t)|^2 dt$$

even function

$$= \int_{0}^{\infty} e^{-2at} dt = \frac{1}{2a} < \infty$$

Hence, the signal is an energy signal.

2. For a system defined by $y(t) = T[x(t)]$, determine if it is additive where the system T is specified by $y(t) = x(t) + 3$. **(5 Points)**

Answer

$T[x_1(t) + x_2(t)] = (x_1(t) + x_2(t)) + 3$

$T[x_1(t)] + T[x_2(t)] = (x_1(t) + 3) + (x_2(t) + 3)$

$= x_1(t) + x_2(t) + 6$

$\therefore (x_1(t) + x_2(t)) + 3 \neq x_1(t) + x_2(t) + 6$,

it is not additive.

3. Is $y(t) = x(t) + 3$ homogenous? **(5 Points)**

Answer

$T[\alpha x(t)] = \alpha x(t) + 3$ — (I)

$\alpha T[x(t)] = \alpha(x(t) + 3)$

$= \alpha x(t) + 3\alpha$ — (II)

$\therefore$ R.H.S. of Equation (I) and Equation (II) are not same

they do not satisfy homogeneity.